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List of Abbreviations

Abbreviation	Meaning
3GPP	3rd Generation Partnership Project
ACK	Acknowledgement
ARQ	Automatic Repeat Request
BCCH	Broadcast Control Channel
BCH	Broadcast Channel
CA	Carrier Aggregation
CCCH	Common Control Channel
CP	Cyclic Prefix
CQI	Channel Quality Indicator
CS	Circuit-Switched
DCCH	Dedicated Control Channel
DCI	Downlink Control Information
DFDMA	Distributed Frequency Division Multiple Access
DFT	Discrete Fourier Transform
DL	Downlink
DL-SCH	Downlink Shared Channel
DRX	Discontinuous Reception
DTCH	Dedicated Traffic Channel
eNodeB	Evolved Node B
EPC	Evolved Packet Core
EPS	Evolved Packet System
E-UTRAN	Evolved UMTS Terrestrial Radio Access Network
FDMA	Frequency Division Multiple Access
FFT	Fast Fourier Transform
GPRS	General Packet Radio Service
GTP	GPRS Tunneling Protocol
GTP-C	GPRS Tunneling Protocol Control Plane
GTP-U	GPRS Tunneling Protocol User Plane
HARQ	Hybrid Automatic Repeat Request
HSS	Home Subscriber Server
ICIC	Inter-Cell Interference Coordination
IDFT	Inverse Discrete Fourier Transform
IFFT	Inverse Fast Fourier Transform
IP	Internet Protocol
ISI	Inter-Symbol Interference
LFDMA	Localized Frequency Division Multiple Access
LTE	Long Term Evolution
LTE-A	Long-Term Evolution Advanced

Abbreviation	Meaning
MAC	Medium Access Control
MBMS	Multimedia Broadcast Multicast Service
MCCH	Multicast Control Channel
MCH	Multicast Channel
MIB	Master Information Block
MIMO	Multiple-Input Multiple-Output
MME	Mobility Management Entity
MTCH	Multicast Traffic Channel
NACK	Negative Acknowledgement
OFDM	Orthogonal Frequency Division Multiplexing
OFDMA	Orthogonal Frequency Division Multiple Access
PA	Power Amplifier
PAPR	Peak-to-Average Power Ratio
PBCH	Physical Broadcast Channel
PCCH	Paging Control Channel
PCFICH	Physical Control Format Indicator Channel
PCH	Paging Channel
PDCP	Packet Data Convergence Protocol
PDCCH	Physical Downlink Control Channel
PDSCH	Physical Downlink Shared Channel
P-GW	Packet Data Network Gateway
PHICH	Physical Hybrid ARQ Indicator Channel
PLMN	Public Land Mobile Network
PMCH	Physical Multicast Channel
PRACH	Physical Random Access Channel
PS	Packet-Switched
PUCCH	Physical Uplink Control Channel
PUSCH	Physical Uplink Shared Channel
QAM	Quadrature Amplitude Modulation
QoS	Quality of Service
QPSK	Quadrature Phase-Shift Keying
RACH	Random Access Channel
RB	Resource Block
RE	Resource Element
RLC	Radio Link Control
RNC	Radio Network Controller
S1AP	S1 Application Protocol
SC-FDMA	Single-Carrier Frequency Division Multiple Access
SCTP	Stream Control Transmission Protocol
S-GW	Serving Gateway

Abbreviation	Meaning
SIB	System Information Block
SR	Scheduling Request
TB	Transport Block
TDMA	Time Division Multiple Access
TTI	Transmission Time Interval
UCI	Uplink Control Information
UDP	User Datagram Protocol
UE	User Equipment
UL	Uplink
UL-SCH	Uplink Shared Channel
USIM	Universal Subscriber Identity Module
X2AP	X2 Application Protocol

Abstract

This report examines the fundamental architecture of the LTE-Advanced Radio Interface, focusing on the highly structured data flow between the Evolved Packet System (EPS) core and the mobile User Equipment (UE). It systematically analyzes the E-UTRAN logical interfaces, the rigorous hierarchical channel structure (Logical, Transport, and Physical), and the advanced multiple access schemes that govern spectrum sharing. By dissecting Orthogonal Frequency Division Multiple Access (OFDMA) in the downlink and Single-Carrier Frequency Division Multiple Access (SC-FDMA) in the uplink, this report highlights how the radio interface successfully supports advanced 4G features such as Carrier Aggregation and dynamic link adaptation while maintaining strict spectral efficiency and low latency constraints.

1 Introduction & EPS Logical Interfaces

1.1 Introduction to the Evolved Packet System (EPS)

The Evolved Packet System (EPS) is the core network architecture introduced by the 3rd Generation Partnership Project (3GPP) starting from Release 8 to support high-speed, low-latency IP connectivity for LTE and LTE-Advanced networks [3].

Unlike legacy 2G/3G mobile networks that relied on a dual-domain infrastructure—Circuit-Switched (CS) for voice and Packet-Switched (PS) for data—the EPS adopts a flat, all-IP packet-switched network topology. This flat architecture eliminates the centralized Radio Network Controller (RNC) to minimize signaling overhead, reduce latency, and optimize user-plane throughput [3].

The EPS is logically divided into three primary segments [3]:

1. **User Equipment (UE):** The mobile terminal containing the Universal Subscriber Identity Module (USIM) that interfaces with the radio access network over the air interface.
2. **Evolved UTRAN (E-UTRAN):** The access network layer composed entirely of evolved NodeBs (eNodeBs). The eNodeBs host the physical layer, MAC, RLC, and PDCP protocols, making direct decisions regarding radio resource management.
3. **Evolved Packet Core (EPC):** The IP-based core backbone responsible for high-level tasks including subscriber authentication, mobility management, dynamic policy enforcement (QoS), and packet routing to external networks.

1.2 EPS Logical Interfaces

To coordinate mobility management, session establishment, and end-to-end data transfer, the nodes within the EPS communicate via highly standardized logical interfaces. These interfaces are strictly divided into the Control Plane (for signaling, routing policies, and security) and the User Plane (for routing actual IP data traffic).

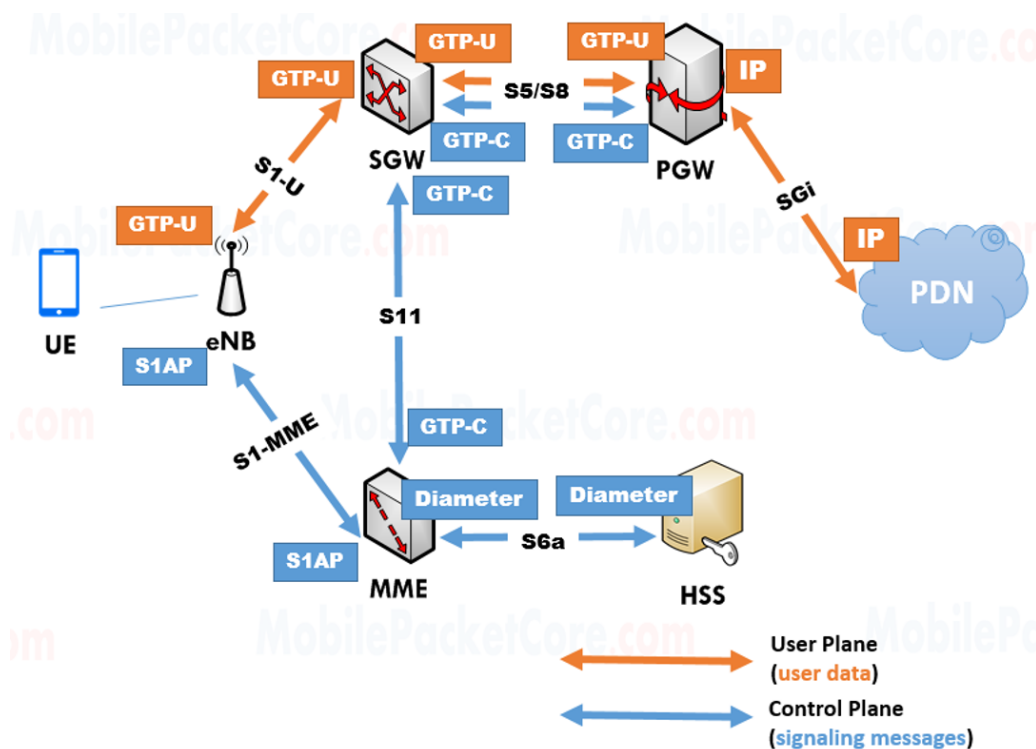


Figure 1: Logical Interfaces of LTE-Advanced [4].

As illustrated in the architecture diagram above, the user plane protocols rely on GTP-U, while the control plane protocols rely on S1AP, GTP-C, and Diameter to carry out the signaling procedures described below [4]:

1.2.1 The S1 Interface

The S1 interface serves as the primary bridge between the E-UTRAN and the EPC. It is logically bifurcated into:

- **S1-MME (Control Plane):** Connects the eNodeB to the Mobility Management Entity (MME). It runs the S1 Application Protocol (S1AP) over the Stream Control Transmission Protocol (SCTP). It handles context setup, paging, and handover signaling.
- **S1-U (User Plane):** Connects the eNodeB to the Serving Gateway (S-GW). It utilizes the GPRS Tunneling Protocol User Plane (GTP-U) over UDP/IP to securely transport user-plane packets.

1.2.2 The X2 Interface

The X2 interface provides a direct point-to-point connection between neighboring eNodeBs, optimizing user mobility.

- **X2-CP (Control Plane):** Employs the X2AP protocol over SCTP for fast handover coordination, cell overload status, and inter-cell interference coordination (ICIC).
- **X2-U (User Plane):** Uses GTP-U over UDP to temporarily forward cached downlink packets from a source eNodeB to a target eNodeB during handovers, preventing packet loss.

1.2.3 The S11 Interface

- Connects the MME directly to the S-GW. It runs the GTP-v2C protocol to establish, modify, and terminate GTP tunnels (bearers) whenever a UE shifts between active and idle states.

1.2.4 The S6a Interface

- Connects the MME to the Home Subscriber Server (HSS). Operating over the Diameter protocol, it is used to download subscriber data, authenticate identities, and verify roaming rights.

1.2.5 The S5/S8 Interfaces

- **S5 Interface:** Links the S-GW to the Packet Data Network Gateway (P-GW) within the same Public Land Mobile Network (PLMN).
- **S8 Interface:** The roaming counterpart of S5, connecting an S-GW in a visited PLMN to the home P-GW. Both interfaces utilize GTP-U and GTP-C.

1.2.6 The Gx Interface

- Connects the P-GW to the Policy and Charging Rules Function (PCRF). It uses the Diameter protocol to enforce dynamic Quality of Service (QoS) profiles and charging models per active user session.

2 Channel Hierarchy Overview & Logical Channels

2.1 Channel Hierarchy Overview

The LTE-Advanced (LTE-A) radio interface architecture is designed to handle complex data and signaling efficiently. To achieve this, the system organizes data flow through a strict hierarchical structure, separating the *type* of data being transmitted from the *method* of its transmission. This hierarchy operates across the Medium Access Control (MAC) and Physical (PHY) layers and is divided into three distinct channel types [6]. Figure 2 illustrates this conceptual mapping:

- **Logical Channels:** Defined by *what* type of information is being transmitted. These channels represent the services offered by the MAC layer to the Radio Link Control (RLC) layer.
- **Transport Channels:** Defined by *how* and with what characteristics the information is transmitted over the radio interface. These channels represent the services offered by the Physical layer to the MAC layer.
- **Physical Channels:** Represent the actual physical resources (time-frequency resource elements) used for transmission over the air interface.

LTE-Advanced (LTE-A) Overall Channel Hierarchy Architecture

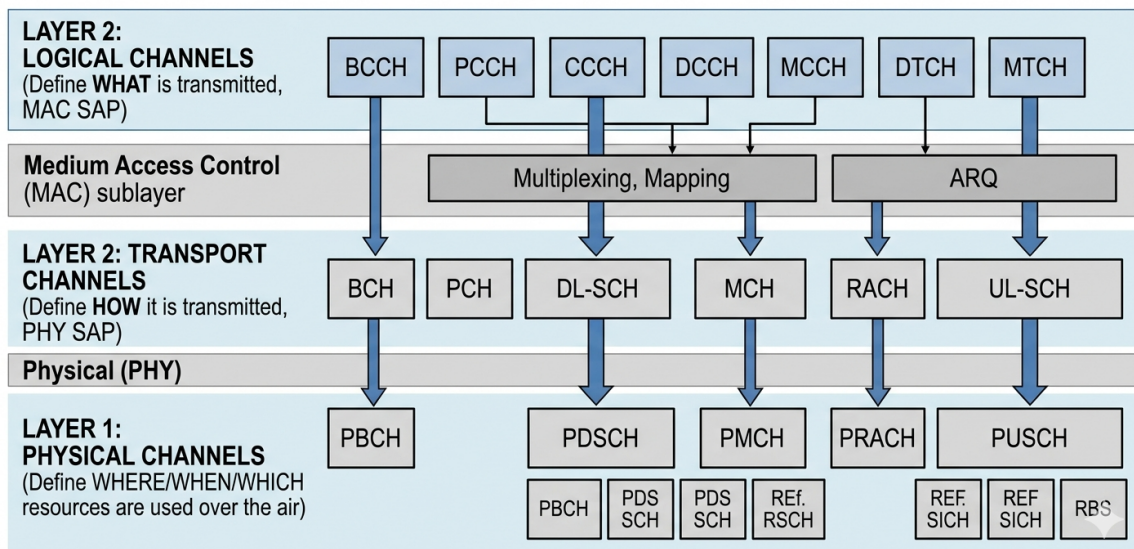


Figure 2: LTE-Advanced (LTE-A) Overall Channel Hierarchy Architecture.

This section focuses exclusively on the highest level of this hierarchy: the **Logical Channels**.

2.2 Logical Channels

Logical channels are characterized by the nature of the data payload they carry. They are strictly structured based on their function in either the Control Plane (signaling) or the User Plane (application data). This functional division is illustrated in Figure 3:

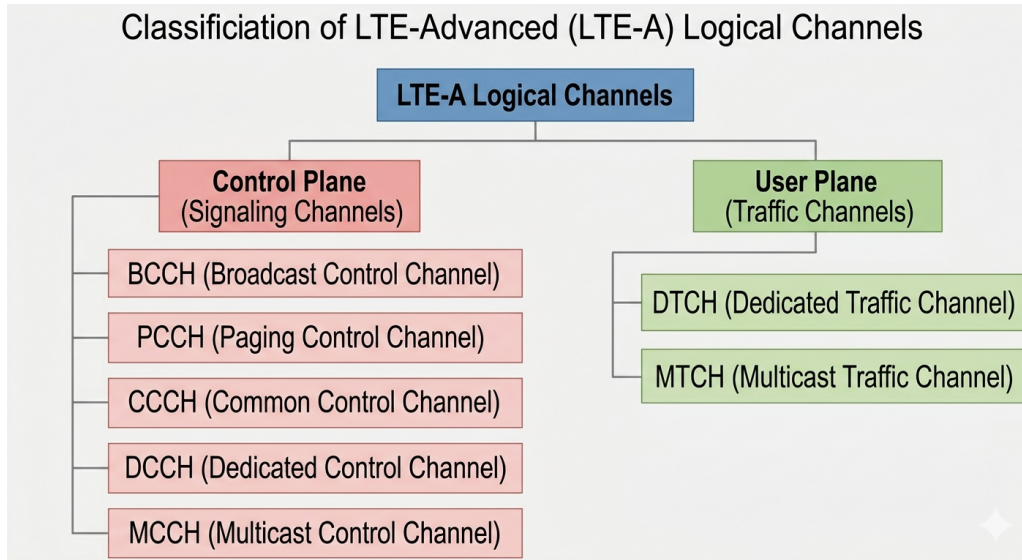


Figure 3: Classification of LTE-Advanced (LTE-A) Logical Channels.

2.3 Logical Control Channels

Control channels are essential for establishing, maintaining, and releasing connections, as well as broadcasting network parameters [7]. The LTE-A standard defines the following logical control channels:

- **BCCH (Broadcast Control Channel):** A downlink channel used to broadcast system control information to all User Equipment (UE) within a specific cell. This includes the Master Information Block (MIB) and System Information Blocks (SIBs), which provide UEs with the necessary parameters to synchronize with and access the network.
- **PCCH (Paging Control Channel):** A downlink channel used to page UEs whose exact cell location is unknown to the network (typically UEs in `RRC_IDLE` mode). The PCCH is also used to notify UEs of incoming calls, SMS, or changes to the system information broadcasted on the BCCH.
- **CCCH (Common Control Channel):** A bidirectional channel used for transmitting control information between the network and UEs that do not currently have a dedicated Radio Resource Control (RRC) connection. It is most heavily utilized during the Random Access Procedure (e.g., initial network attachment or recovering from a radio link failure).
- **DCCH (Dedicated Control Channel):** A point-to-point, bidirectional channel that transmits dedicated control information between a specific UE and the network. The DCCH is established only after the UE has successfully completed the RRC connection setup process, handling tasks like measurement reporting and handover signaling.
- **MCCH (Multicast Control Channel):** A point-to-multipoint downlink channel used for transmitting control information related to the Multimedia Broadcast Multicast Service (MBMS). It signals the parameters required for UEs to receive multicast data traffic.

2.4 Logical Traffic Channels

Once signaling is handled and a connection is established, the network utilizes traffic channels to transfer the actual user payload [6]. There are two types of logical traffic channels in LTE-A:

- **DTCH (Dedicated Traffic Channel):** A point-to-point, bidirectional channel dedicated to a single UE for the transfer of user data. All standard internet browsing, voice-over-LTE (VoLTE) calls, and application data transfers occur over the DTCH. A UE can have multiple DTCHs active simultaneously for different data bearers.
- **MTCH (Multicast Traffic Channel):** A point-to-multipoint downlink channel used to transmit MBMS user data from the network to multiple UEs simultaneously. This is highly efficient for broadcasting live video feeds or large-scale emergency alerts where the same data payload is required by many users within a cell.

3 Transport Channels (Downlink & Uplink)

Within the LTE-Advanced protocol stack, information flows through a strictly hierarchical structure. While Logical Channels define *what* type of data is being transferred (e.g., dedicated user traffic vs. broadcast control data), **Transport Channels** dictate *how* and with what characteristics this data is transferred over the radio interface [1]. They act as the critical bridge between the Medium Access Control (MAC) layer and the Physical (PHY) layer.

The MAC layer is responsible for multiplexing the various Logical Channels into discrete payload units called Transport Blocks (TBs). These blocks are then mapped onto the corresponding Transport Channels, passing down to the PHY layer once every Transmission Time Interval (TTI), which is defined as 1 millisecond in LTE-A [1].

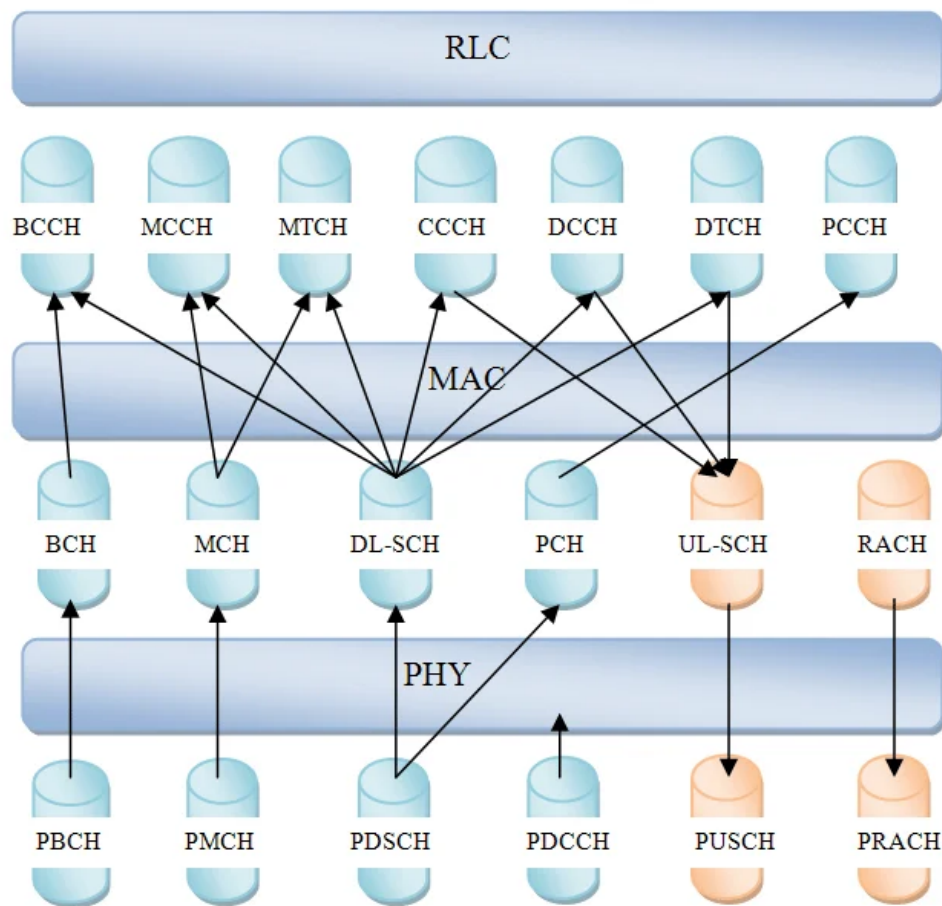


Figure 4: Mapping of Logical, Transport, and Physical Channels [1].

As illustrated in Figure 4, the network does not map data on a simple one-to-one basis. Instead, the MAC layer dynamically aggregates channels based on their functional purpose [1]. To simplify the architecture from previous 3G standards, LTE-A uses exclusively shared and broadcast channels, eliminating dedicated transport channels entirely. The transport channels can be grouped based on their primary functions:

Downlink Transport Channels

- **Downlink Shared Channel (DL-SCH):** This is the absolute core of the LTE-A downlink. It multiplexes user payload (DTCH) and dedicated control signaling (DCCH

and CCCH). The DL-SCH supports highly dynamic link adaptation, meaning the eNodeB changes the modulation format and transmit power on a sub-millisecond basis depending on the UE's reported channel quality. Crucially, the DL-SCH also supports Hybrid ARQ (HARQ) for rapid error correction, and in LTE-Advanced, it can process multiple Transport Blocks across different frequency bands simultaneously to support Carrier Aggregation (CA) [1, 2].

- **Broadcast Channel (BCH):** Maps directly from the BCCH to deliver the Master Information Block (MIB). It has a rigid, fixed transport format because every UE in the cell must be able to decode it to synchronize with the network [1].
- **Paging Channel (PCH):** Associated with the PCCH, this channel is used to wake UEs from idle modes. It supports Discontinuous Reception (DRX), a critical feature that allows the mobile device's receiver to power down and only wake up to check the PCH, saving immense battery life.
- **Multicast Channel (MCH):** Used exclusively to support Multimedia Broadcast Multicast Services (MBMS) by mapping the MCCH and MTCH, allowing the network to stream the same media to multiple UEs simultaneously [1].

Uplink Transport Channels

- **Uplink Shared Channel (UL-SCH):** The exact uplink equivalent of the DL-SCH. It carries all user data and dedicated control information from the UE to the eNodeB. Like its downlink counterpart, it supports HARQ and dynamic link adaptation [1].
- **Random Access Channel (RACH):** This channel is unique because it is not mapped to any logical channel. It is used strictly for initial network access. When a UE powers on or wakes from idle mode, it transmits a random access preamble on the RACH to request uplink scheduling resources [1].

Table 2: Functional Summary of LTE-A Transport Channels [1].

Channel	Direction	Primary Function and Characteristics
DL-SCH	Downlink	Carries user data; supports dynamic link adaptation, Carrier Aggregation, and HARQ.
BCH	Downlink	Broadcasts essential system information (MIB) with a fixed, highly robust format.
PCH	Downlink	Transmits paging messages; supports UE power-saving (DRX).
MCH	Downlink	Delivers synchronized broadcast/multicast multimedia streams.
UL-SCH	Uplink	Carries user data; supports dynamic link adaptation and HARQ.
RACH	Uplink	Transmits initial access preambles for network synchronization.

As detailed in Table 2, these transport channels manage the physical characteristics of the data payload. Once the MAC layer has processed the Transport Blocks on these respective channels, they are forwarded to the Physical layer, where they are mapped onto waveforms (like the PDSCH and PUSCH) for actual radio transmission over the air interface [1].

4 Physical Channels & Channel Mapping

In the LTE-Advanced radio interface, Physical Channels represent the specific time-frequency resources (Resource Blocks) in the frame structure that carry modulated symbols over the air interface. They act as the physical pipeline for the higher-layer transport channels [5].

4.1 Downlink Physical Channels

- **PBCH (Physical Broadcast Channel):** Carries the Master Information Block (MIB). It transmits essential parameters required for the UE to sync with the cell, such as system bandwidth and system frame number.
- **PDSCH (Physical Downlink Shared Channel):** The main data-bearing channel. It carries user payload (DL-SCH) and System Information Blocks (SIBs).
- **PDCCH (Physical Downlink Control Channel):** Transmits Downlink Control Information (DCI), which provides uplink resource allocation grants and downlink scheduling parameters.
- **PCFICH (Physical Control Format Indicator Channel):** Signals whether the PDCCH control region occupies 1, 2, or 3 OFDM symbols at the beginning of the subframe.
- **PHICH (Physical Hybrid ARQ Indicator Channel):** Carries the downlink HARQ ACK/NACK responses corresponding to uplink transmissions from the UE.
- **PMCH (Physical Multicast Channel):** Dedicated to Multimedia Broadcast Multicast Services (MBMS) for point-to-multipoint streaming over a single frequency network.

4.2 Uplink Physical Channels

- **PUSCH (Physical Uplink Shared Channel):** The primary channel used by the UE to transmit uplink user data.
- **PUCCH (Physical Uplink Control Channel):** Transmits Uplink Control Information (UCI) when there is no user data. This includes Channel Quality Indicators (CQI), Scheduling Requests (SR), and HARQ ACK/NACK for downlink packets.
- **PRACH (Physical Random Access Channel):** Carries the random access preamble sequence from the UE, enabling initial access and connection setup.

4.3 Channel Mapping

To transmit data, upper-layer information flows from Logical Channels (what is being transmitted) down to Transport Channels (how it is transmitted over the MAC layer), which are then mapped to Physical Channels (the exact physical resources) [5].

4.3.1 Downlink Mapping

- **BCCH (Logical)** maps to **BCH (Transport)**, which maps to **PBCH (Physical)**.
- **BCCH / PCCH / CCCH / DCCH / DTCH (Logical)** map to **DL-SCH / PCH (Transport)**, which map to **PDSCH (Physical)**.
- **MCCH / MTCH (Logical)** map to **MCH (Transport)**, which map to **PMCH (Physical)**.

4.3.2 Uplink Mapping

- **CCCH / DCCH / DTCH (Logical)** map to **UL-SCH (Transport)**, which map to **PUSCH (Physical)**.
- **RACH (Transport)** maps directly to **PRACH (Physical)** without an upper-layer logical mapping.

5 Downlink Multiple Access – OFDMA

A central design challenge in any cellular network is to allow many User Equipments (UEs) to share the limited radio spectrum of a single cell without mutually destructive interference. To address this, LTE-Advanced adopts **Orthogonal Frequency Division Multiple Access (OFDMA)** as the downlink multiple-access scheme, replacing the CDMA-based access used in previous 3G generations [3]. OFDMA combines the multipath robustness of Orthogonal Frequency Division Multiplexing (OFDM) with a flexible time–frequency resource allocation mechanism, making it well-suited to the high-throughput, packet-switched nature of the Evolved Packet System (EPS).

5.1 Principle of Operation

OFDMA is built on top of OFDM. In conventional OFDM, the available system bandwidth is split into a large number of narrowband *subcarriers* that are mathematically orthogonal: the spectral peak of each subcarrier coincides exactly with the spectral nulls of all the others, so the subcarriers can overlap in frequency without interfering with one another [8]. In LTE-Advanced, the subcarrier spacing is fixed at $\Delta f = 15$ kHz, which yields an OFDM symbol duration of $T_{\text{sym}} = 1/\Delta f \approx 66.7 \mu\text{s}$ [8].

OFDMA extends this idea to a multi-user environment. Instead of allocating the entire OFDM symbol to a single user (as in plain OFDM), the eNodeB *partitions* the subcarriers among multiple UEs, so that different users are served simultaneously on disjoint groups of subcarriers within the same OFDM symbol. Each transmitted subcarrier is independently modulated using QPSK, 16-QAM, or 64-QAM, with the modulation order chosen dynamically based on the channel conditions reported by each UE [8].

The transmitter chain at the eNodeB can be summarized as follows:

1. Incoming bits from the transport channels are mapped to complex modulation symbols (QPSK / 16-QAM / 64-QAM).
2. The scheduler assigns each UE's symbols to a specific set of subcarriers (i.e. specific Resource Blocks).
3. An Inverse Fast Fourier Transform (IFFT) converts the frequency-domain symbols into a time-domain OFDM waveform.
4. A *Cyclic Prefix* (CP) is appended to each OFDM symbol to absorb multipath delay spread and eliminate Inter-Symbol Interference (ISI) between consecutive symbols.

At the receiver, the UE removes the CP, performs an FFT, and extracts only the subcarriers that the eNodeB allocated to it.

5.2 The Resource Block as the Scheduling Unit

To make scheduling tractable, LTE-Advanced does not allocate individual subcarriers; instead, it groups them into **Resource Blocks (RBs)**. A Resource Block consists of 12 consecutive subcarriers in the frequency domain (a span of $12 \times 15 \text{ kHz} = 180 \text{ kHz}$) and one 0.5 ms slot in the time domain, which corresponds to 7 OFDM symbols when the normal cyclic prefix is used [8]. The smallest addressable unit, a single subcarrier within a single OFDM symbol, is called a *Resource Element* (RE).

The eNodeB scheduler allocates RBs to UEs once every Transmission Time Interval (TTI) of 1 ms. The total number of available RBs depends on the channel bandwidth: for example, a 5 MHz carrier provides 25 RBs, while a 20 MHz carrier provides 100 RBs [8]. Because allocation is performed on this fine-grained two-dimensional grid, the eNodeB can exploit the fact that fading is frequency-selective: different UEs typically experience their best channel conditions on different subcarriers, and the scheduler can hand each UE the RBs on which *its* channel is strongest. This is known as *frequency-domain multi-user diversity* and is one of the main reasons OFDMA is more spectrally efficient than the single-carrier schemes used in earlier generations [3].

5.3 Advantages of OFDMA in the Downlink

The choice of OFDMA for the LTE-Advanced downlink is motivated by several practical advantages [3, 8]:

- **Robustness to multipath fading:** Because each subcarrier is narrowband, the channel seen by any one subcarrier is approximately flat. Combined with the cyclic prefix, this allows simple one-tap frequency-domain equalisation at the UE, even in highly dispersive urban environments.
- **High and flexible spectral efficiency:** The eNodeB can dynamically adapt the modulation and coding scheme on a per-RB, per-TTI basis according to each UE's reported Channel Quality Indicator (CQI), so users with good channels are served at high rates while users at the cell edge fall back to more robust modulation.
- **Flexible bandwidth scaling:** The same OFDMA structure supports the full set of LTE-Advanced channel bandwidths (1.4, 3, 5, 10, 15, and 20 MHz) simply by changing the number of allocated RBs, which also makes OFDMA a natural fit for Carrier Aggregation across multiple component carriers.
- **Straightforward MIMO integration:** Because equalisation is performed in the frequency domain on a per-subcarrier basis, spatial multiplexing and beamforming techniques (Multiple-Input Multiple-Output, MIMO) can be applied independently on each subcarrier, which is essential for reaching the multi-Gbps peak rates targeted by LTE-Advanced.
- **Fine-grained multi-user scheduling:** The two-dimensional time–frequency grid lets the eNodeB exploit multi-user diversity and enforce per-user Quality-of-Service constraints with much higher resolution than CDMA-based 3G systems [3].

5.4 Limitations

The main drawback of OFDMA is its high **Peak-to-Average Power Ratio (PAPR)**: because the transmitted signal is the sum of many independently modulated subcarriers, occasional constructive addition produces large signal peaks. A high PAPR forces the power amplifier to operate with a significant back-off, which reduces power-amplifier efficiency [8]. In the downlink, where the eNodeB has access to mains power and bulky amplifiers, this is acceptable. In the uplink, however, the UE is battery-powered and a high PAPR would shorten battery life unacceptably; this is precisely why LTE-Advanced uses SC-FDMA (discussed in a later section) rather than OFDMA on the uplink.

6 Uplink: SC-FDMA (Single-Carrier Frequency Division Multiple Access)

SC-FDMA is used in the LTE-Advanced uplink (from UE to eNodeB) as the counterpart to OFDMA in the downlink. The two figures below (Figure 5 and Figure 6) illustrate the SC-FDMA block diagram and its subcarrier mapping strategies respectively.

- **Principle:** While based on OFDM principles, SC-FDMA introduces an additional Discrete Fourier Transform (DFT) processing step before the standard OFDM IFFT modulation [11, 10]. The UE's data symbols are first spread across the assigned subcarriers by the DFT block. This precoding step results in the transmitted signal having characteristics closer to a single-carrier signal [12]. Subcarriers allocated to a UE can be contiguous (**Localized FDMA – LFDMA**, primarily used in LTE-Advanced) or distributed (**Distributed FDMA – DFDMA**) [10, 11].

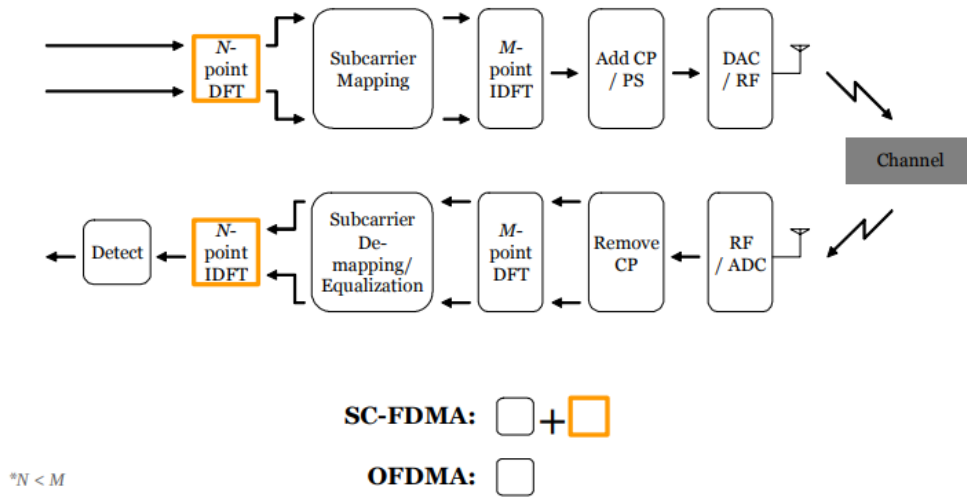


Figure 5: SC-FDMA Block Diagram [12].

As shown in Figure 5, the key distinction from OFDMA is the additional DFT precoding block at the transmitter, which spreads the data symbols across the assigned subcarriers before the standard IFFT step, resulting in a lower PAPR signal.

- **Rationale for the Uplink:** The primary reason for choosing SC-FDMA over OFDMA in the uplink is its lower **Peak-to-Average Power Ratio (PAPR)** [11, 10]. OFDMA signals can exhibit high peaks in power due to the coherent summation of multiple subcarriers, requiring expensive, linear, and less power-efficient amplifiers in the transmitter. For battery-powered UEs, using SC-FDMA allows for simpler, more power-efficient amplifiers, conserving battery life while maintaining coverage [12, 13].

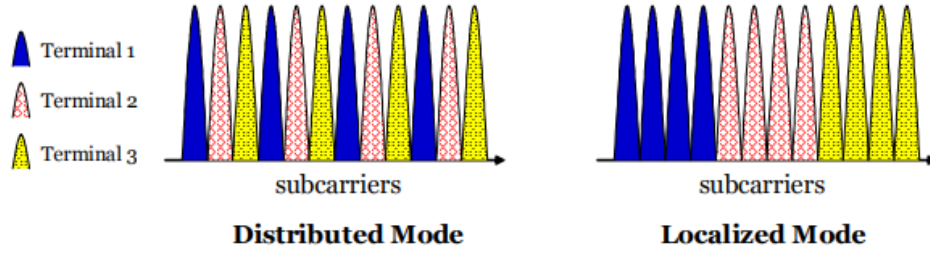


Figure 6: LFDMA vs. DFDMA Subcarrier Mapping [12].

As shown in Figure 6, in LFDMA the assigned subcarriers form a contiguous block within the bandwidth, while in DFDMA they are spread with equal spacing across the full spectrum.

- **LTE-Advanced Enhancements:** Include “Clustered SC-FDMA,” which allows a UE to be allocated multiple non-contiguous clusters of subcarriers, providing more scheduling flexibility compared to pure LFDMA while still maintaining lower PAPR than OFDMA [9]. Simultaneous transmission of data (on PUSCH) and control information (on PUCCH) is also supported [9, 14].

7 Conclusion

The LTE-Advanced standard achieves its performance objectives through the synergistic operation of a well-defined network architecture, a structured channel hierarchy, and sophisticated multiple access techniques.

The logical interfaces defined in the 3GPP standards provide the essential framework for communication and interoperability between the various nodes within the LTE-Advanced Evolved Packet System. Interfaces like S1, X2, S6a, S11, and S5/S8 enable critical functions ranging from user plane data transport and control signalling to mobility management and policy enforcement [9, 14].

The hierarchical channel structure, comprising Logical, Transport, and Physical channels, ensures the organised and efficient flow of different types of information over the radio interface. This layered approach allows for flexible mapping and processing of data and control signals based on their nature and the requirements of the physical transmission [9].

Complementing the network architecture and channel structure, the multiple access techniques are fundamental to radio interface performance. OFDMA in the downlink offers high spectral efficiency, robustness, and flexibility suitable for the high-power eNodeB transmitter. SC-FDMA in the uplink provides similar performance benefits but with the crucial advantage of lower PAPR, making it ideal for power-constrained user equipment [12, 10, 13]. Together, these well-defined interfaces, structured channels, and sophisticated multiple access schemes are cornerstone technologies enabling LTE-Advanced to deliver high-speed, low-latency mobile broadband services.

References

- [1] E. Dahlman, S. Parkvall, and J. Sköld, “The LTE Radio Interface,” in *4G LTE/LTE-Advanced for Mobile Broadband*, 2nd ed. Oxford, UK: Academic Press, 2014, ch. 14, pp. 451–491.
- [2] G. Yuan *et al.*, “Carrier Aggregation for LTE-Advanced Mobile Communication Systems,” *IEEE Communications Magazine*, vol. 48, no. 2, pp. 88–93, Feb. 2010.
- [3] G. A. Abed, M. Ismail, and K. Jumari, “The evolution to 4G cellular systems: Architecture and key features of LTE-Advanced networks,” *IRACST - International Journal of Computer Networks and Wireless Communications (IJCNC)*, vol. 2, no. 1, pp. 21–26, 2012.
- [4] “LTE Interfaces,” *Mobile Packet Core*. Available at: <https://mobilepacketcore.com/lte-interfaces/>
- [5] “Logical and Physical Channels,” *Ques10*, 2019. Available at: <https://www.ques10.com/p/48287/logical-and-physical-channels/>
- [6] 3GPP, “Evolved Universal Terrestrial Radio Access (E-UTRA); Medium Access Control (MAC) protocol specification,” TS 36.321.
- [7] 3GPP, “Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification,” TS 36.331.
- [8] “4G LTE Modulation: OFDM, OFDMA, SC-FDMA,” *Electronics Notes*. Available: <https://www.electronics-notes.com/articles/connectivity/4g-lte-long-term-evolution/ofdm-ofdma-scfdma-modulation.php>
- [9] Rohde & Schwarz, “LTE / LTE-Advanced Fundamentals,” Application Note. Available: https://www.rohde-schwarz.com/us/applications/lte-lte-advanced-fundamentals-application-note_56280-15658.html
- [10] A. C. Grammenos and I. Darwazeh, “SC-FDMA and OFDMA: The two competing technologies for LTE,” *Proc. 14th International OFDM Workshop*, Hamburg, 2010. Available: <https://discovery.ucl.ac.uk/id/eprint/1529403/>
- [11] R. Jaiswal and A. Jain, “Multiple access technology in LTE,” *International Refereed Journal of Engineering and Science (IRJES)*, vol. 3, no. 1, pp. 32–38, 2014. Available: <http://www.irjes.com/Papers/vol3-issue1/Vesion%201/F03013238.pdf>
- [12] H. G. Myung, “Introduction to Single Carrier FDMA,” *Proc. 15th European Signal Processing Conference (EUSIPCO)*, Poznan, Poland, 2007. Available: <https://www.eurasip.org/Proceedings/Eusipco/Eusipco2007/Papers/d3l-a01.pdf>
- [13] Commsbrief, “Difference between OFDM, OFDMA and SC-FDMA in 4G LTE.” Available: <https://commsbrief.com/difference-between-ofdm-ofdma-and-scfdma-in-4g-lte/>
- [14] 3GPP, “TS 36.211: E-UTRA; Physical channels and modulation (Release 10),” Tech. Spec., 2011. Available: https://www.3gpp.org/ftp/Specs/archive/36_series/36.211/